

Manual

Shop Floor Control BDE-FST 8.2

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1 Overview: Shop Floor Control

Purpose

This component combines different applications that support planning in the area of shop floor data collection.

You use the function package if:

- You want to run simple sequencing for your machines and workplaces without reference to schedules or without considering capacities,
- You need a list specifying when which order must be set up and whether in the process a setup change is made,
- You need an overview indicating on which day how much material is needed.

Integration

Sequencing, done in tabular form, specifies the processing sequence that is also integrated in the sequencing list of the terminal.

The data displayed in the setup change list and in the material requirements overview are based on detailed schedule planning performed either in the ERP system or in HYDRA Shop Floor Scheduling.

Features

- Order sequencing
 - Tabular order sequencing for planning (sequence planning) the pool of orders for machine groups and workplace groups without accounting for any capacity restrictions
 - Simple drag and drop assignment of operations to machines and workplaces
- Setup change list
 - Setup change list displaying orders / operations that must be converted during the period selected. Differentiation by tool change, color change and product change. Display of any other relevant information (setup time, tool number, workforce requirements, articles etc.)
- Material requirements list
 - Material requirements list with material requirements calculated for the selected machine, cost center or machine group and the time period selected including daily totals. Display of any other relevant information about operations, articles to be produced and materials

2 Order sequencing

Overview

HYDRA menu	Production control -> Production support -> Order sequencing
FEDRA menu	Detailed scheduling → Planning → Order sequencing
Transaction code	orseq
Function authorization	orseq

Available user fields		
Where	Object type/user field key	Source (type)
Table <i>Pool of groups</i>	AUNR/SYSTEM	Order (MF-D)
Table <i>Pool of groups</i>	AGNR/SYSTEM	Operation (MF-D)
Table <i>Pool of workplaces</i>	AUNR/SYSTEM	Order (MF-D)
Table <i>Pool of workplaces</i>	AGNR/SYSTEM	Operation (MF-D)
How can I configure user fields? Which user field types are available?		

Purpose

Order sequencing is a function of production management. Use the application to simply sequence workplaces without taking into account any restrictions. You can use *drag & drop* to perform planning in the order sequencing function.

This application is an ideal tool for planners, shift foremen and supervisors. You are provided with simple means to plan the sequence of operations on machines.

Integration

The application provides functions for the following tasks:

- Display operations that are stored in relation to a group within the pool.
- Display operations, which have already been planned on workplaces.
- Plan an operation onto a workplace/machine.
- Define a processing sequence ("sequencing") that specifies how the operations to be produced are displayed in the sequencing list on the terminal
- Re-plan an operation on another workplace or group
- Merge operations
- Split operations
- Lock/unlock operations

Restrictions



You can plan workplaces either only via the *order sequencing* or the *graphic planning* (shop floor scheduling). We recommend to plan individual workplace/machine groups either only by using the *order sequencing* or the *graphic planning* function.

Selection criteria

In general, the function only selects operations that are currently

- not logged on and
- not finished

(The system selects operations with an operation status assigned to the control indicators V, U, S).

Use selection parameters to restrict the pool of groups. But selection criteria do not affect the pool of workplaces/machines.

The application provides the following selection criteria:

Group

Selects operations relating to a machine group

Order type

Selects the order type

Order

Selects an order

Article/item

Selects an article

Article name

Selects an article name/designation

Controlling preceding OP

Selects operations subject to the status of the preceding OP

Order index from ... to

Selects the order index

Priority from ... to

Selects the priority

Scheduled start from ... to ...

Selects the scheduled start of operations

Scheduled end from ... to ...

Selects the scheduled end of operations

Earliest start from ... to ...

Selects the earliest start of operations

Earliest end from ... to ...

Selects the earliest end of operations

Check responsibility area

Use this selection option to specify if you want to check the responsibility area of

- the *workplace* or
- the *operation/order*

in order to display data. You need the function authorization *chkresp* to select this option.

Detail application *Pool of groups*

The "pool of groups" table shows detailed information on individual [operations](#).



Please note that the responsibility area is not checked for operations that are included in the pool of groups.

Detail application *Workplaces*

This detail dialog shows the workplaces that are assigned to the currently selected group. If you select a workplace in this dialog, the "pool of workplaces" dialog shows the operations that are planned on this workplace.

The detail application only shows the workplaces meeting the following criteria:

- The planning function is set to the option "planning in order sequencing"
- The workplace is not locked
- The planner is authorized for the responsibility area of this workplace

The following columns are available:

Workplace

Unique identification of the workplace/machine according to configuration

Short name

Short name of the workplace according to configuration

Name (designation)

Workplace name according to configuration

Group

Group assigned to the workplace according to configuration

Cost center

Cost center assigned to the workplace according to configuration

Company

Company assigned to the workplace according to configuration

Responsibility area

Responsibility area assigned to the workplace according to configuration

Workplace type

Type of workplace according to configuration:

E = individual workstation; G = group workstation



The table is sorted by workplace. For technical reasons, rearrangement and grouping are not supported.

Detail application *Pool of workplaces*

This detail application shows the [operations](#) that are planned for the currently selected workplace. In this detail application the sort sequence is fixed and cannot be changed.



The system uses a separate, internal algorithm to generate the sort sequence of operations in the pool of workplaces. The result is stored in special database fields. The *pool of workplaces* shows this result in the *time of sorting* column.

The time of sorting does not change if an operation is logged on and interrupted later. The operation lines up in the list of planned operations according to the time of sorting.

Editing functions

Please note that HYDRA does not perform any validation checks as to whether the operation may be produced on the planned workplace.

Proceed as described below to plan operations (detailed planning):

In the *pool of groups* select the operations you want to plan on a workplace. Left click the corresponding entry to select an operation. To select several operations at once, click the first operation, press the shift button, hold it and click the last operation. To select specific operations, press the *Ctrl* button, hold it and click the required operations.

In the *pool of workplaces* select the workplace where you want to plan operations. On the left hand side of the *pool of workplaces* dialog, click the workplace where you want to plan the selected operations.

Transfer the selected operations to the *pool of workplaces* using “drag and drop” Left click an operation selected in the pool of groups, hold the mouse button down and move the mouse pointer to the right hand area of the *pool of workplaces* dialog (“drag”) and then release it (“drop”).

Once you have released the left mouse button, the selected operations disappear in the *pool of groups* and are now visible in the *pool of workplaces*.

If necessary, change the sequence of operations planned on the workplace

You can change the sequence of operations planned on the workplace. To do so, drag the selected operation and drop it above the operation before which you want to plan this selected OP. The sort function of the table is deactivated as this would contradict the principles of sequencing.

The server downloads operations to the terminals (sequencing list). The terminal shows the planned operations in the [terminal's sequencing list](#). The MOC application "order sequencing" specifies the sequence of the operations included in this sequencing list.

Toolbar

General index tab



Order information (function authorization: orin)

This button opens the application [Order information](#).



Order overview (function authorization: orov)

This button opens the application [Order overview](#).

Entry index tab



Log on operation (function authorization: op.logon)

Use the [log on](#) function to log on operations to the system.



Partial confirmation (report part quantities; function authorization: op.partconf)

Use the function [partial confirmation \(report part quantities\)](#) to report part quantities for operations.



Interrupt operation (function authorization: op.interrupt)

Use the function [interrupt operation](#) to interrupt operations.



Log off operation (function authorization: op.logoff)

Use the [log off](#) function to log off operations.

**Terminate operation (function authorization: op.finish)**

Use the [terminate](#) function to log off interrupted or prepared operations.

Other functions index tab**Split operation (function authorization: op.split)**

Use the [split](#) function to split operations.

**Cancel split operation (function authorization: op.splitrelease)**

Use the [dissolve split](#) function to cancel splits.

**Generate merged operation (function authorization: op.colopcreate)**

Use the [generate merged operation](#) function to summarize interrupted or prepared operations in merged operations.

**Cancel merged operation (function authorization: op.coloprelease)**

Use the [cancel merged operation](#) function to cancel merged operations.

3 Setup Change List

Overview

HYDRA menu	Production control --> Planning aid --> Setup change list
FEDRA menu	Detailed Scheduling → Planning → Setup change list
Transaction code	setli
Function authorization	setli

Available user fields		
Where?	Object type/user field key	Source (type)
Setup change list table	AGNR/SYSTEM	Operation (MF-D)
How to configure user fields?		Which user field types are available?

Purpose

The setup change list provides the user with an overview of the machines to be set up next. The operations listed which are planned in next. Setups, due to material, tool or color changes, are marked to improve your overview.

In addition, the setup change list has been designed as preview for (subsequent) shifts, which enables the responsible persons in the shift to know which machines have to be set up next.

Requirements

Operations have to be planned exactly with respect to dates to be able to determine setup changes.

Selection criteria

The application provides the following selection criteria:

Date ... until

Shows all planned setup changes (planned operations) that coincide with the selected date range. The option "consider job end" option (see below) specifies which operations are exactly to be taken into account.

Cost center

Shows the setup changes that are planned on workplaces assigned to the selected cost center.

Workplace

Shows the setup changes that are planned on the selected workplaces.

Group

Shows the setup changes that are planned on the workplaces of the selected groups.

Order category

Shows the setup changes that are planned for orders of the selected order category.

Consider job end

Generally, only planned operations assigned to the control indicators L, S, V or U are taken into account. Subject to the "consider job end" option, the result list and, as a result the table, shows the following operations:

- The "consider job end" option is set: operations the start of which is prior to the selection period and the planned end of which coincides with the selection period are displayed. This applies to planned as well as running operations.
- The "consider job end" option is set: the below-mentioned operations are considered in addition to operations the planned start of which coincides with the selection period:
 - operations the start of which is prior to the selection period and the planned end of which coincides with the selection period
 - operations the start of which is prior to the selection period and with the planned end after the selection period. This applies to planned operations (control indicators S, V, U) as well as running operations (control indicator L).

It can be reasonable NOT to check the "consider job end" option, provided that setup change lists are to be printed for each shift in advance.

Field descriptions

In general, if selected by workplace, group and cost center, only operations are selected that are actually planned on a workplace. Operations that are in the pool of groups are not selected. In addition, only operations are selected that are planned on workplaces for the responsibility area of which the user is authorized.

Remarks on selected columns

Workplace

Shows the workplace where the OP is currently planned for OPs that are not running (status U or V). For running OPs, the workplace is displayed where the OP is currently logged on.

Group

Group to which the workplace is assigned.

Cost center

Cost center to which the workplace is assigned.

Status

Current operation status. Shows the colored LED as well as the status text.

Article

The value is taken from the operation.

←Planned start/planned end

The planned dates can be taken from the order information, tab Detailed Planning, under Planned Dates - Start or End. Note: the planned dates as per planning are used; the remaining run time is not taken into account.

Target duration

The target duration is calculated as follows: $(\text{target quantity} * \text{target cycle} / \text{part} / 1000) * \text{machine efficiency} / 100$ ←←

Tool

The value is taken from the operation.

Storage location

(Original) storage location that is assigned to the tool. Please note: This information is only provided if the HYDRA tool and resource management (HYDRA-WRM) module is in use and edited in this master data.

Current storage location

Storage location that the tool is assigned to on the basis of its logon as a resource of type WNR. The material buffer that is assigned to the workplace in the workplace/resource configuration as the upstream material buffer is used as the storage location.

Note: The assignment is only done in case of using the HYDRA Tool and Resource Management (WRM).

Color

The value is taken from the operation.

Addition

The value is taken from the operation.

Planned start (setup date)

Planned setup date; corresponds to the planned start date of the operation.

Change

See below.

"Symbol"

See below.

Setup time

Total of the values "setup time" and "additional setup time" at the operation.

M/O rel. setup

Machine/operator relation for setup; the value is taken from the operation.

Table setup change, symbol

When setting the value in the Setup Change column, the neighboring operations are evaluated (according to the planned start date).

- If the value in the Tool field changes, the value "Tool" and the symbol are entered. 
- If the value in the Tool field does not change, but the value in the Color field does, then the "Color" value and the icon are entered. 
- If the value in the Tool and Color fields does not change, but the value in the Material field does, then the value "Material" and the symbol are entered. 
- If the value in the Tool, Color and Material fields does not change, but the value in the Article field does, then the value "Article" and the icon will be entered.  The symbol for "Article" and "Material" is the same.



These values remain even if the user chooses another table sorting that does not correspond to the sequence of the planned dates.

The fields described above are the fields that are directly adjacent to the operation.

Toolbar

When you call a function or target application, the parameters of the table are always transferred. For this reason, always select an entry to call an application.



Order information (function authorization: orin)

Use this button to call the application [Order information](#).



Order overview (function authorization: orov)

Use this button to call the application [Order overview](#).

4 List of Material Requirements

Overview

HYDRA menu	Production control → Planning support → List of material requirements
FEDRA menu	Detailed scheduling → Planning → Material requirements list
Transaction code	mrqli
Function authorization	mrqli

Purpose

The *List of material requirements* is an application of the production management. The application provides a preview of the materials required at the machines. This application therefore addresses employees in production who are responsible for material staging at machines in due time.

Depending on how planning is organized, material dispatchers who are responsible for calling off material from suppliers with framework contracts at short-term can also use this application.

This application therefore clearly focuses on short-term activities.

Requirements

Operations must be scheduled accurately in order to calculate material requirements.

Selection criteria

The system selects the operations if the following conditions are true:

- Operations with the status "prepared" or "interrupted" are planned for the (selected) machines. Their planned start time is in the future and between the specified *Point in time* (in the future) and the end time after the specified *No. of days*.
- Operations with status "running" are logged on to the (selected) machines.

Only those operations are selected whose component list includes at least one material component.

Prepared and interrupted operations are not selected if their start time is in the past. In this case, the system does not know when these operations will be active again.

The system neither selects operations where the option *External processing OP* is enabled.

If you select by *Workplace* or *Group*, the system only selects operations that are planned for a specific workplace. Operations that are integrated in the pool of groups (backlog for machine group) are not selected. In general, the system identifies the responsibility area the user is authorized for and only selects the operations that are planned for workplaces of this responsibility area.

The application provides the following selection criteria:

Order type

The application displays the material requirements of the selected order type.

Order

The application displays the material requirements of the selected order.

Operation

The application displays the material requirements of the selected operation.

Workplace ... to ...

The application displays the material requirements of the selected workplace.

Cost center

The application displays the material requirements of the selected cost center.

Group ... to ...

The application displays the material requirements of the selected group.

Date and time

The application displays the material requirements from the specified point in time.

Number of days

Use the *No. of days* option to specify the time horizon (in calendar days). The material requirements for the specified time are then displayed (maximum 365 days).

Field descriptions

Planned start

The planned dates of the operation are defined in the *Order information*, tab *Dates*.

Planned end

The planned end is calculated using the planned start + remaining run time (using the shift calendar) to calculate the material requirements. The calculated planned end is not displayed.

For running OPs, the field *Planned start* shows the first logon time.

Remaining run time

Shows the remaining production time. The remaining run time is a calculated value. The system uses different parameters and a formula to calculate the value. The formula is stored for the operation in field *Formula RRT1*.

Remaining run time*

The remaining production time from start time of the material requirements calculation. In the graph, this time is identified as RRT*.

Material requirements

The quantity of the single component that is required on the relevant shift day. A preview of up to a maximum of 365 calendar days is possible.

Detail application

The detail application *Material requirements* provides an overview of the materials required at the machines. The material requirements of each material are exactly calculated for each day using the shift calendar. The detail application therefore shows the distribution of the material requirements during the runtime of the OP(s).

"For each day" means: the calculation is made for each shift day, i.e. the day starts at the beginning of shift 1 and ends with the end of the last shift (e.g. shift 3).

The calculation of the material requirements per shift day is described in section "Calculation of the material requirements".



The application only shows the material components if the following is true:

$$\left(\sum_1^{\text{Number of days}} \text{Material requirements (per shift day)} \right) > 0$$

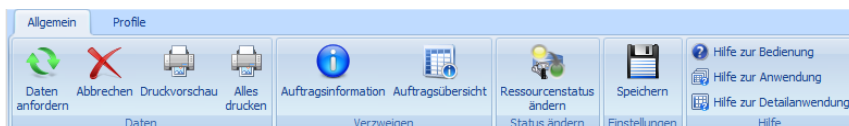
This evaluation does not integrate *External processing OPs*.

Because of different shift models, the shift times (shift start/end) can be different for each workplace. The application does not integrate these different shift times when displaying the separate shift days.

The evaluation does not integrate operations that are logged on at the same time. Consequently, the target duration is not calculated proportionately. Recommendation: do not log on an operation in parallel to different workplaces, but split the operation instead.

Toolbar

When you call a function or target application, the parameters included in the table are used. For this reason, always select an entry to call an application.



Order information

Use this button to call the application [Order information](#).



Order overview

Use this button to call the application [Order overview](#).



Modify resource status (function authorization: op.resstatchg)

Function to [change the resource status of an operation](#)

Calculation of material requirements

When the system calculates when and how much material must be available at the workplace, the system uses the planned start date. The material must be available at this start date. The material must be available during setup to check machine settings, for example.

Using the selection criteria specified, the system calculates the material requirements for **each operation** as follows:

1. Calculation of the **remaining quantity of the operation**:

$$\text{remaining quantity} = \text{target quantity } (P)_{OP} - \text{yield } (P)_{OP}$$

2. Calculation of the remaining run time of the operation using the *Formula RRT1* stored for the operation. The performance level of the workplace where the operation is planned is used to calculate the remaining run time.

Example: If the remaining run time is 300 minutes (calculated using the remaining run time formula), the remaining run time relevant for further calculation is

- with a performance level of 100%: $300 / 100 * 100 = 300$ minutes
- with a performance level of 80%: $300 / 80 * 100 = 375$ minutes
- with a performance level of 120%: $300 / 120 * 100 = 250$ minutes.

A performance level of 0 is internally regarded as 100%.

3. Identification of the **calculated time per piece**:

$$\text{calculated time per piece} = \frac{\text{remaining run time of operation}}{\text{remaining quantity of operation}}$$

4. Calculation of the **time per shift day for the workplace** where the operation is planned or currently logged on. This calculated time is also referred to as **target time**.

If a planned year model is stored for the workplace (go to: Workplace/resource configuration > tab Workplace configuration > field Planned year model), this planned year model is used to identify the shift days. Otherwise the shift model used for data collection is used (go to: Workplace/resource configuration < tab Workplace configuration > field Year model).

5. Calculation of the quantity to be produced on each shift day. This calculated quantity is also referred to as **target quantity**.

$$\text{target quantity (per shift day)} = \frac{\text{target time}}{\text{calculated time per piece}}$$

6. Calculation of **material requirements for each shift day** and for each material component assigned to the operation:

$$\text{material requirements (per shift day)} = \text{input quantity of material} * \text{target quantity (per shift day)}$$

Step 4 to 6 are carried out for each shift day where the operation is planned. For each shift day that follows, the system reduces the remaining quantity by the target quantity of the previous day.